

ForgeCloud Frontend Design

React control plane, pages, components, UX and API integration

1. Frontend Responsibility

The React frontend is the ForgeCloud control plane presented to developers and administrators. It provides deployment forms, application lifecycle screens, dashboards, monitoring views, logs, infrastructure status and administrative controls.

2. Frontend Stack

Technology	Purpose
React.js	Component-based control panel
JavaScript	Application logic
REST API	Communication with FastAPI
Charts/UI components	Monitoring and operational visualization

3. Pages

- /login
- /dashboard
- /applications
- /applications/create
- /applications/:id
- /applications/:id/deployments
- /applications/:id/infrastructure
- /monitoring
- /logs
- /infrastructure
- /settings
- /admin/users
- /admin/audit

4. Frontend Architecture

```
React App
|
+-- Authentication Context
|
+-- Layout
|   +-- Sidebar
|   +-- Header
|
+-- Pages
|   +-- Dashboard
|   +-- Applications
```

- +-- Deployment
- +-- Monitoring
- +-- Logs
- +-- Infrastructure
- +-- Admin

- +-- Components
 - +-- StatusBadge
 - +-- MetricCard
 - +-- DeploymentTable
 - +-- ApplicationForm
 - +-- Charts

- +-- Services
 - +-- authService
 - +-- applicationService
 - +-- deploymentService
 - +-- monitoringService

5. Dashboard Design

ForgeCloud Dashboard

Applications

12

Running

9

Failed

1

CPU Usage

68%

Memory

59%

P95 Latency

184 ms

Recent Deployments

frontend-api

v1.4.2

SUCCESS

payment-api

v2.1.0

SUCCESS

user-service

v1.2.1

ROLLED BACK

System Health

API

HEALTHY

Database

HEALTHY

EKS

HEALTHY

Registry

HEALTHY

6. Application Details Screen

Application: inventory-api

Status: RUNNING

Version: v1.4.2

Replicas: 3/3

CPU: 42%

Memory: 51%

[Deploy]

[Rollback]

[Restart]

Deployment History

v1.4.2

SUCCESS

v1.4.1

SUCCESS

v1.4.0

FAILED -> ROLLBACK

Metrics: CPU | Memory | Requests | Latency

Logs: timestamp | level | message

7. UX Requirements

- Responsive layout.
- Clear deployment status indicators.
- Loading, empty and error states.
- Confirmation for destructive actions such as rollback or deletion.

- No fake operational data once real APIs are available.
- Consistent navigation and terminology.
- Role-aware controls so users see only actions they are allowed to perform.

8. Frontend API Contract

```
GET    /api/applications
POST   /api/applications
GET    /api/applications/{id}
PUT    /api/applications/{id}
DELETE /api/applications/{id}

POST   /api/applications/{id}/deploy
POST   /api/applications/{id}/rollback

GET    /api/applications/{id}/deployments
GET    /api/metrics
GET    /api/logs
GET    /api/health
```